AI-Powered Smart University Assistant

Final Project Report

Course: Artificial Intelligence / Machine Learning
Assignment: Group Practical Assignment â€” AI-Powered Smart University Assistant
Total Marks: 100

Title Page

Project Title: AI-Powered Smart University Assistant
Team Members:

Name	Student ID	Role
Team Member 1	[ID]	Data Preprocessing & Feature Engineering
Team Member 2	[ID]	Machine Learning Model Development
Team Member 3	[ID]	Chatbot & Recommendation System
Team Member 4	[ID]	Dashboard Development & Explainable AI

Submission Date: [Date]
Generative AI Declaration: GitHub Copilot and Antigravity AI coding assistant were used to assist in code development. All AI outputs were reviewed, verified, and adapted by team members.

Abstract

This report presents the design and development of an AI-Powered Smart University Assistant, a prototype system that integrates machine learning, natural language processing, a recommendation engine, and explainable AI into a single interactive web application. Using the Open University Learning Analytics Dataset (OULAD) containing 32,593 student records, the system predicts academic risk, recommends personalized learning resources, answers university-related queries through an intelligent chatbot, and provides transparent explanations for every prediction. The system achieves an F1-score of approximately 0.75 on the full-semester risk model and demonstrates that early identification of at-risk students using only 30 days of data is feasible, albeit with reduced accuracy.

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1. Introduction

The rapid digitization of educational environments has produced vast quantities of student behavioral data that institutions have yet to leverage effectively. Academic advisors continue to identify struggling students through manual observation, often too late in the semester for meaningful intervention. This project addresses this gap by developing an intelligent university assistant that automatically analyzes student learning data and provides actionable insights to both students and educators.

The system integrates five areas of artificial intelligence:

1. Machine Learning for academic risk classification
2. Natural Language Processing for FAQ retrieval
3. A rule-based recommendation system for learning resources
4. Explainable AI through SHAP analysis
5. An interactive Streamlit dashboard for intelligent decision support

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2. Problem Statement

A university aims to improve student support services using artificial intelligence. The key challenges are:

- Academic advisors identify weak students manually, often mid or late semester
- Students struggle to find relevant course information
- Learning resources are not personalized to individual performance
- Prediction systems are "black boxes" that advisors do not trust

Stakeholders:

- **Students** â€” primary beneficiaries of risk warnings and resource recommendations
- **Academic Advisors** â€” users of the risk prediction and SHAP explanation tools
- **University Administration** â€” interested in systemic patterns and at-risk rates
- **IT Department** â€” responsible for maintaining the deployed application

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3. Assignment Objectives

After completing this system, the team demonstrated ability to:

- Analyze the OULAD real-world educational dataset
- Clean, transform, and integrate multiple data files
- Apply exploratory data analysis across 8+ visualizations
- Develop and compare three machine learning classification models
- Build an NLP-based question-answering chatbot using Sentence Transformers
- Design a rule-based recommendation system linked to course modules
- Apply SHAP explainable AI techniques globally and locally
- Develop a fully integrated Streamlit web application
- Evaluate ethical and privacy implications of educational AI

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4. Related Work

Educational data mining and learning analytics have emerged as significant research areas. Romero and Ventura (2010) surveyed early applications of data mining to e-learning, establishing foundational patterns for predicting student performance. Aljohani et al. (2019) demonstrated that VLE interaction data from OULAD is particularly predictive of student outcomes, with ensemble methods outperforming traditional classifiers. More recently, Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), which has become the standard approach for explaining tree-based model predictions in high-stakes domains such as education.

5. Dataset Description

5.1 Primary Dataset: OULAD

The Open University Learning Analytics Dataset (OULAD) provides anonymized student records from the UK's Open University.

File	Records	Description
studentInfo.csv	32,593	Demographics, registration, final result
studentAssessment.csv	173,912	Assessment scores per student
assessments.csv	206	Assessment metadata
studentVle.csv	~10M	VLE click interactions
studentRegistration.csv	32,593	Registration and withdrawal dates

Target Variable: Binary classification â€" `At_Risk = 1` if final_result is 'Fail' or 'Withdrawn', `At_Risk = 0` if 'Pass' or 'Distinction'.

Class Distribution:

- At Risk: 17,208 (52.8%)
- Not At Risk: 15,385 (47.2%)

5.2 FAQ Dataset

A custom `university_faq.csv` file was created containing 100 verified question-answer pairs across 10 categories: Admissions, Fee and Finance, Course Registration, Attendance, Examinations, Scholarships, Internships, Final-Year Projects, Library Services, and Campus Facilities.

5.3 Learning Resources Dataset

A `learning_resources.csv` file was created with 200 resources across all 7 course modules (AAAâ€"GGG) and 3 difficulty levels, with real URLs linking to Scikit-Learn documentation, HuggingFace NLP courses, IBM, and other educational platforms.

6. Data Preprocessing

6.1 Data Cleaning

- `imd_band` NaN values (approximately 1%) were filled with the category 'Missing'
- `studentAssessment.score` NaN values were filled with 0 (students who did not score)
- No duplicate student records were identified
- `date_unregistration` NaN values were coded as 999 (indicating no withdrawal)

6.2 Feature Engineering

Seven features were engineered from multiple OULAD files:

Feature	Source	Description
total_vle_clicks	studentVle	Total number of learning platform interactions
active_learning_days	studentVle	Number of unique days with VLE activity
avg_clicks_per_active_day	Derived	Engagement intensity metric
assessments_attempted	studentAssessment	Number of assessments submitted
average_assessment_score	studentAssessment	Mean assessment score
avg_submission_delay	studentAssessment + assessments	Average days before/after deadline
avg_registration_delay	studentRegistration	Registration timing relative to course start

6.3 Categorical Encoding

One-Hot Encoding was applied to categorical features: `gender`, `region`, `highest_education`, `imd_band`, `age_band`, `disability`. All numerical features were standardized using StandardScaler within sklearn Pipeline objects.

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7. Exploratory Data Analysis

Eight visualizations were produced, revealing the following key patterns:

1. **Result Distribution:** Pass (37.9%), Withdrawn (31.2%), Fail (21.6%), Distinction (9.3%)
2. **Gender:** Male students have a marginally higher at-risk rate than female students
3. **Education Level:** Students with lower prior qualifications show higher at-risk rates
4. **Assessment Scores:** At-risk students average ~38% vs ~82% for not-at-risk students
5. **VLE Activity:** At-risk students generate significantly fewer VLE clicks on average
6. **Engagement vs Score:** Strong positive correlation between engagement and performance
7. **Correlation Matrix:** `average_assessment_score` has the strongest negative correlation with `At_Risk` (-0.62)
8. **Class Distribution:** Near-balanced classes (52.8% vs 47.2%) â€” minimal imbalance correction needed

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8. Machine Learning Models

8.1 Model Architecture

Three algorithms were trained and compared using an sklearn Pipeline:

Model	Type	Imbalance Strategy
Logistic Regression	Linear	class_weight = 'balanced'
Random Forest	Ensemble (Bagging)	class_weight = 'balanced'
XGBoost	Ensemble (Boosting)	scale_pos_weight

8.2 Training Procedure

- **Train/Test Split:** 80/20 with `stratify=y` (random_state = 42)
- **Cross-Validation:** 5-Fold StratifiedKFold on training set
- **Hyperparameter Tuning:** GridSearchCV (3-fold) for Random Forest and XGBoost

8.3 Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.672	0.683	0.710	0.697	0.738
Random Forest (Tuned)	0.724	0.731	0.759	0.745	0.792
XGBoost (Tuned)	0.739	0.740	0.771	0.755	0.810

Final Model: XGBoost (Tuned) selected based on highest F1-score and ROC-AUC.

8.4 Why Recall Matters

In academic risk prediction, a False Negative (missing an at-risk student) has greater consequences than a False Positive (unnecessary advisor check-in). Therefore, the model selection prioritizes **Recall** and **F1-score** over raw accuracy.

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9. Early Warning Prediction

An early warning model was trained using only VLE and assessment data from the **first 30 days** of the course. This simulates predicting risk before most assessments are completed.

Metric	Full Semester	Early Warning (30-day)
F1-Score	~0.755	~0.68
Recall	~0.771	~0.71
ROC-AUC	~0.810	~0.73

Finding: The early warning model can identify approximately 71% of at-risk students using only the first 30 days of data â€” sufficient for meaningful early intervention while accepting a slight reduction in accuracy.

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10. Chatbot Design

The University FAQ Chatbot uses a semantic retrieval approach:

1. **Text Preprocessing:** Lowercase conversion, punctuation removal
2. **Embedding Model:** `all-MiniLM-L6-v2` from Sentence Transformers
3. **Retrieval Method:** Cosine similarity between user query embedding and FAQ embeddings
4. **Confidence Threshold:** 0.5 â€” queries below this threshold trigger the fallback message
5. **Fallback Message:** *"I could not find a sufficiently reliable answer. Please contact the relevant university office or rephrase your question."*

6. **Unanswered Logging:** All fallback queries are logged to `data/unanswered_questions.txt`

Chatbot Evaluation (30 Test Cases)

Category	Questions	Correct	Accuracy
FAQ-similar	15	14	93.3%
Paraphrased	10	8	80.0%
Out-of-scope	5	5 (fallback)	100%
Overall	**30**	**27**	**90.0%**

Average similarity score on correct answers: **0.74**

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11. Recommendation System

The recommendation system uses **Rule-Based Recommendation (Option A)**:

Rule	Condition	Difficulty	Resource Type
1	High risk + score < 50%	Beginner	Tutorial/Video
2	High/Medium risk + low engagement	Beginner	Short Video
3	Medium risk + score 50â€“70%	Intermediate	Practice Exercise
4	Low risk + score â‰¥ 70%	Advanced	Research Paper
5	Default	Intermediate	Article

Resources are filtered by **both the student's course module** (AAAâ€“GGG) and recommended difficulty, ensuring relevance. Real URLs from Scikit-Learn, HuggingFace, IBM, and MachineLearningMastery are provided.

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12. Explainable AI

Global Explanation (SHAP)

The SHAP TreeExplainer generates a summary plot showing the global importance of each feature across all predictions. The top 5 predictors are:

1. Average Assessment Score â€” strongest negative predictor of risk
2. Total VLE Clicks â€” higher engagement = lower risk
3. Active Learning Days â€” consistency matters
4. Number of Previous Attempts â€” repeat attempts correlate with risk
5. Average Submission Delay â€” late submissions are a risk signal

Local Explanation

For each individual student, a SHAP waterfall plot is generated showing which features pushed the prediction toward or away from At-Risk. A human-readable text explanation is also displayed in the dashboard.

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13. Application Development

The application was built with Streamlit and includes 6 pages:

Page	Key Features
Home	Project overview, dataset stats, team info, core capabilities
Student Dashboard	Student profile, assessment score, VLE clicks, risk category
Risk Prediction	Select student, generate ML prediction, probability, SHAP explanation
Learning Recommendations	Course-specific resources, difficulty-matched, real URLs
University Chatbot	Semantic search, confidence score, category display, fallback
Analytics	Real data charts (result/gender/module distributions), model comparison

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14. Results and Discussion

The system successfully demonstrates practical AI integration for educational support. XGBoost achieved the best balance of precision and recall. The chatbot performs well on expected questions but predictably struggles with highly paraphrased or ambiguous queries. The recommendation system appropriately tailors suggestions to both student performance and course module.

A key limitation is that the recommendation rules are manually defined rather than learned from data â€” a hybrid or collaborative filtering approach would improve personalization in a production system.

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15. Ethical Considerations

Student Privacy

All student data uses anonymized IDs (no real names or personally identifiable information). In a production deployment, GDPR and local data protection laws must be followed. Explicit informed consent must be obtained before collecting or processing student data.

Algorithmic Bias

The model was trained on students from the UK Open University. Predictions may not generalize to students from different cultural, economic, or educational backgrounds. The `imd_band` (deprivation index) feature is used for prediction, which raises concerns about reinforcing socioeconomic disadvantage.

Fairness

Gender is included as a feature. While it may improve predictive accuracy, it risks encoding historical biases. Future work should evaluate model fairness across demographic groups.

Misclassification Risk

False negatives (missing at-risk students) and false positives (unnecessary interventions) both have real costs. The system must be positioned as a **decision-support tool** â€” final academic decisions must always involve human judgment.

Human Oversight

Academic advisors should review all AI recommendations before taking action. The SHAP explanations are designed to facilitate this oversight by making model reasoning transparent and understandable to non-technical users.

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16. Limitations

1. The chatbot's knowledge is limited to the university FAQ dataset â€" it cannot answer general academic questions
2. The recommendation system uses fixed rules rather than learned preferences
3. The early warning model has reduced accuracy due to limited data in the first 30 days
4. The system has not been tested with real students in a live university environment
5. Model drift (performance degradation over time) is not currently monitored

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17. Conclusion

This project successfully developed an integrated AI University Assistant that demonstrates the practical application of five areas of artificial intelligence. The system is capable of predicting academic risk with ~75% F1-score, answering 90% of university FAQ queries correctly, and generating course-specific learning recommendations with transparent SHAP explanations. The Streamlit dashboard provides an accessible interface for both students and academic advisors.

Future work should focus on: (1) replacing rule-based recommendations with a content-based or collaborative filtering approach, (2) deploying the system with real student data under proper privacy governance, and (3) implementing continuous model monitoring to detect performance degradation over time.

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18. References

1. Romero, C., & Ventura, S. (2010). Educational Data Mining: A Review of the State of the Art. *IEEE Transactions on Systems, Man, and Cybernetics*, 40(6), 601â€“618.
2. Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics Dataset. *Scientific Data*, 4, 170171.
3. Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NIPS 2017*.
4. Reimers, N., & Gurevych, I. (2019). Sentence-BERT. *EMNLP 2019*.
5. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD 2016*.

Appendix A: Project Folder Structure

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smart-university-assistant/
â”œâ”€â”€ data/
â”‚   â”œâ”€â”€ raw/            (OULAD files)
â”‚   â”œâ”€â”€ processed/      (student_features.csv, early_warning_fea
â”‚   â”œâ”€â”€ university_faq.csv
â”‚   â””â”€â”€ learning_resources.csv
â”œâ”€â”€ notebooks/
â”‚   â”œâ”€â”€ 01_data_understanding.ipynb
â”‚   â”œâ”€â”€ 02_data_preprocessing.ipynb
â”‚   â”œâ”€â”€ 03_exploratory_analysis.ipynb
â”‚   â”œâ”€â”€ 04_model_training.ipynb
â”‚   â”œâ”€â”€ 05_model_evaluation.ipynb
â”‚   â””â”€â”€ 06_explainable_ai.ipynb
â”œâ”€â”€ models/
â”‚   â”œâ”€â”€ risk_prediction_model.pkl
â”‚   â””â”€â”€ early_risk_prediction_model.pkl
â”œâ”€â”€ src/
â”‚   â”œâ”€â”€ preprocessing.py
â”‚   â”œâ”€â”€ prediction.py
â”‚   â”œâ”€â”€ chatbot.py
â”‚   â”œâ”€â”€ recommender.py
â”‚   â””â”€â”€ explainability.py
â”œâ”€â”€ app/app.py
â”œâ”€â”€ screenshots/
â”œâ”€â”€ report/
â”œâ”€â”€ README.md
â””â”€â”€ requirements.txt
```

Appendix B: Individual Contribution Statement

Team Member	Contribution	Percentage
[Member 1]	Data collection, preprocessing, EDA notebooks	25%
[Member 2]	Model training, evaluation, hyperparameter tuning	25%
[Member 3]	Chatbot development, recommendation system	25%
[Member 4]	Dashboard development, SHAP integration, report	25%

All team members reviewed and verified each other's work.